

Conservation, Energy Geometry, and Reversibility

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Definition: First integral

Let $U \subseteq \mathbb{R}^2$ be open and let $\mathbf{F} : U \rightarrow \mathbb{R}^2$ be continuously differentiable. A continuously differentiable function $E : U \rightarrow \mathbb{R}$ is a **first integral** of $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$ if

$$\nabla E(\mathbf{z}) \cdot \mathbf{F}(\mathbf{z}) = 0$$

for every $\mathbf{z} \in U$. Equivalently, $E(\mathbf{z}(t))$ is constant along every solution.

Definition: Conservative system

A system on an open set U is **conservative** if it has a first integral that is not constant on any nonempty open subset of U .

Question: Why is mechanical energy conserved?

Let $I \subseteq \mathbb{R}$ be an open interval, let $V \in C^2(I)$, let $m > 0$, and suppose $x(t) \in I$ satisfies

$$m\ddot{x} = F(x), \quad F(x) = -V'(x).$$

Find a first integral of the planar system obtained with $y = \dot{x}$.

Solution. The planar system is

$$\dot{x} = y, \quad \dot{y} = -\frac{1}{m}V'(x).$$

Multiplying the second-order equation by $\dot{x}$ gives

$$m\dot{x}\ddot{x} + V'(x)\dot{x} = \frac{d}{dt} \left(\frac{1}{2}m\dot{x}^2 + V(x) \right) = 0.$$

Thus

$$E(x, y) = \frac{1}{2}my^2 + V(x)$$

is conserved. Directly,

$$\mathbf{G}(x, y) = \left(y, -\frac{1}{m}V'(x) \right), \quad \nabla E \cdot \mathbf{G} = (V'(x), my) \cdot \left(y, -\frac{1}{m}V'(x) \right) = 0.$$

Proposition: A conservative system has no attracting equilibrium

A conservative system has no attracting equilibrium.

Proof. Let E be a first integral that is not constant on any nonempty open set. Suppose $\mathbf{z}^*$ is attracting. Its basin contains a nonempty open neighborhood of $\mathbf{z}^*$. For any $\mathbf{z}_0$ in the basin, conservation and continuity give

$$E(\mathbf{z}_0) = E(\Phi_t(\mathbf{z}_0)) \longrightarrow E(\mathbf{z}^*) \quad (t \rightarrow \infty).$$

Hence $E(\mathbf{z}_0) = E(\mathbf{z}^*)$ for every $\mathbf{z}_0$ in the basin. Thus E is constant on a nonempty open set, contradicting the definition of a conservative system.

Visualization: Potential and phase-plane energy level

Let $I \subseteq \mathbb{R}$ be an interval, let $m > 0$, let $V \in C^2(I)$, and consider

$$\dot{x} = y, \quad \dot{y} = -\frac{1}{m}V'(x), \quad E(x, y) = \frac{1}{2}my^2 + V(x).$$

For

$$E(x, y) = \frac{1}{2}my^2 + V(x) = h,$$

the two velocity branches are

$$y = \pm \sqrt{\frac{2}{m}(h - V(x))}.$$

Therefore

potential condition	phase-plane meaning
$V(x) < h$	two states with velocities $\pm y $
$V(x) = h, V'(x) \neq 0$	$y = 0$; non-equilibrium turning point
$V(x) = h, V'(x) = 0$	equilibrium
$V(x) > h$	no state on the energy level
$V'(x^*) = 0, V''(x^*) > 0$	nearby oval levels about $(x^*, 0)$
$V'(x^*) = 0, V''(x^*) < 0$	saddle energy at $(x^*, 0)$

Since $y = \dot{x}$, the upper branch moves right and the lower branch moves left.

Definition: Regular level

Let $U \subseteq \mathbb{R}^2$ be open, let $E \in C^1(U)$, and let $h \in \mathbb{R}$. The level set

$$\{p \in U : E(p) = h\}$$

is **regular** if

$$\nabla E(p) \neq \mathbf{0}$$

at every point p in that level set.

Recall: Hessian matrix

For $E \in C^2(U)$ on an open set $U \subseteq \mathbb{R}^2$, its **Hessian matrix** is

$$D^2E = \begin{pmatrix} E_{xx} & E_{xy} \\ E_{yx} & E_{yy} \end{pmatrix}.$$

The matrix is symmetric because $E_{xy} = E_{yx}$.

Recall: Definite symmetric matrix

A real symmetric matrix H is **positive definite** if $\mathbf{v}^\top H \mathbf{v} > 0$ for every nonzero vector $\mathbf{v}$. It is **negative definite** if the inequality is reversed. Here $\top$ denotes transpose.

Lemma: Definite Hessian gives closed nearby levels

Let E be twice continuously differentiable near $\mathbf{z}^* \in \mathbb{R}^2$, suppose

$$\nabla E(\mathbf{z}^*) = \mathbf{0},$$

and suppose $D^2E(\mathbf{z}^*)$ is positive definite. Then $\mathbf{z}^*$ is a strict local minimum, and for all sufficiently small $c > 0$ the nearby component of

$$E(\mathbf{z}) = E(\mathbf{z}^*) + c$$

is a regular simple closed curve surrounding $\mathbf{z}^*$. For a negative definite Hessian, the same conclusion holds with $c < 0$.

Proof. Assume the Hessian is positive definite. By continuity, there are a radius $R > 0$ and a constant $m > 0$ such that

$$m\|\mathbf{v}\|^2 \leq \mathbf{v}^\top D^2E(\mathbf{z})\mathbf{v}$$

for every $\mathbf{v} \in \mathbb{R}^2$ whenever $\|\mathbf{z} - \mathbf{z}^*\| \leq R$. For each unit vector $\mathbf{e} \in \mathbb{R}^2$, define

$$q_{\mathbf{e}}(r) = E(\mathbf{z}^* + r\mathbf{e}) - E(\mathbf{z}^*).$$

Because $\nabla E(\mathbf{z}^*) = \mathbf{0}$,

$$q'_{\mathbf{e}}(r) = \int_0^r \mathbf{e}^\top D^2E(\mathbf{z}^* + s\mathbf{e})\mathbf{e} \, ds,$$

so, for $0 < r \leq R$,

$$q'_{\mathbf{e}}(r) \geq mr > 0, \quad q_{\mathbf{e}}(r) \geq \frac{m}{2}r^2.$$

The second inequality makes $\mathbf{z}^*$ a strict local minimum. The first makes the energy strictly increase along each ray. Thus, for every $0 < c < mR^2/2$ and every unit vector $\mathbf{e}$, there is a unique radius $r_c(\mathbf{e}) \in (0, R)$ satisfying

$$E(\mathbf{z}^* + r_c(\mathbf{e})\mathbf{e}) = E(\mathbf{z}^*) + c.$$

The implicit-function theorem makes r_c continuously differentiable in the direction $\mathbf{e}$. As $\mathbf{e}$ runs around the unit circle,

$$\mathbf{e} \mapsto \mathbf{z}^* + r_c(\mathbf{e})\mathbf{e}$$

parametrizes a simple closed curve surrounding $\mathbf{z}^*$. Moreover, $q'_{\mathbf{e}}(r_c(\mathbf{e})) > 0$, so ∇E does not vanish on this curve; the level is regular. Replacing E by $-E$ proves the negative-definite case. $\square$

Proposition: Regular closed energy levels are periodic orbits

Let E be a continuously differentiable first integral of a continuously differentiable planar vector field $\mathbf{F}$. Let C be a regular simple closed component of a level set of E . If C contains no equilibrium, then C is a periodic orbit.

Proof. Conservation keeps trajectories on C , and compactness permits continuation for all time. Since C contains no equilibrium, the speed $\|\mathbf{F}\|$ has a positive minimum m on C . The direction of travel cannot reverse without the vector field vanishing. If L is the length of C , one circuit takes time

$$T = \int_C \frac{ds}{\|\mathbf{F}\|} \leq \frac{L}{m} < \infty.$$

Uniqueness makes the returning trajectory periodic.

Definition: Nonlinear center

An isolated equilibrium $\mathbf{z}^*$ of a planar system is a **nonlinear center** if every sufficiently nearby non-equilibrium point lies on a periodic orbit that is a simple closed curve surrounding $\mathbf{z}^*$.

Theorem: Definite-Hessian nonlinear-center criterion

Let $\mathbf{F}$ be continuously differentiable near an isolated equilibrium $\mathbf{z}^*$ of a planar system, and let E be a twice continuously differentiable first integral. Suppose

$$\nabla E(\mathbf{z}^*) = \mathbf{0}$$

and $D^2E(\mathbf{z}^*)$ is positive definite or negative definite. Then every non-equilibrium point sufficiently near $\mathbf{z}^*$ lies on a periodic orbit that is a simple closed level curve of E surrounding $\mathbf{z}^*$. Hence $\mathbf{z}^*$ is a nonlinear center.

Proof. Assume the Hessian is positive definite; otherwise replace E by $-E$. Since $\mathbf{z}^*$ is isolated, apply the preceding lemma in a disk containing no other equilibrium. Every non-equilibrium point sufficiently near $\mathbf{z}^*$ lies on one of the lemma's regular simple closed energy curves in that disk. The preceding proposition makes each curve a periodic orbit.

Question: Why is a non-strict local minimum insufficient?

Consider

$$\dot{x} = 0, \quad \dot{y} = x^2 + y^2, \quad E(x, y) = x^2.$$

Verify that the origin is an isolated equilibrium and a local minimum of a first integral that is not constant on any nonempty open set, but is not a center.

Solution. The equilibrium equations are

$$0 = 0, \quad x^2 + y^2 = 0,$$

so the origin is the unique equilibrium. Also,

$$\dot{E} = 2x\dot{x} = 0,$$

Question: Why is a non-strict local minimum insufficient? (continued)

and $E = x^2 \geq 0 = E(0, 0)$. The minimum is not strict, and the nearby levels $x^2 = c$ are pairs of vertical lines rather than closed curves. At every non-equilibrium point,

$$\dot{y} = x^2 + y^2 > 0,$$

so no non-equilibrium trajectory is periodic. The missing hypothesis is the closed-contour condition; a definite Hessian would guarantee it.

Finally, E is not constant on any nonempty open set: every open disk contains a horizontal line segment on which x^2 is not constant.

Definition: Homoclinic trajectory

Let $U \subseteq \mathbb{R}^2$ be open, let $\mathbf{F} : U \rightarrow \mathbb{R}^2$, and let $p \in U$ be an equilibrium of $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$. A nonconstant solution $\mathbf{z} : \mathbb{R} \rightarrow U$ is **homoclinic to** p if

$$\lim_{t \rightarrow -\infty} \mathbf{z}(t) = p = \lim_{t \rightarrow \infty} \mathbf{z}(t).$$

Its image is a **homoclinic trajectory** or **homoclinic orbit**.

Question: What is the complete phase portrait of a double well?

Consider

$$\ddot{x} = x - x^3.$$

Find the equilibria, conserved energy, local classifications, and energy-level regimes.

Solution. With $y = \dot{x}$,

$$\dot{x} = y, \quad \dot{y} = x - x^3.$$

Choose

$$V(x) = -\frac{1}{2}x^2 + \frac{1}{4}x^4, \quad E(x, y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4.$$

The equilibria are

$$(0, 0), \quad (1, 0), \quad (-1, 0).$$

The Jacobian is

$$J(x, y) = \begin{pmatrix} 0 & 1 \\ 1 - 3x^2 & 0 \end{pmatrix}.$$

At $(0, 0)$, the eigenvalues are ± 1 , so it is a saddle. At $(\pm 1, 0)$, the eigenvalues are $\pm i\sqrt{2}$, and

$$D^2E(\pm 1, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

is positive definite. Hence $(\pm 1, 0)$ are nonlinear centers.

Their energy is $-1/4$ and the saddle energy is 0. On $E = h$,

$$y = \pm \sqrt{2h + x^2 - \frac{1}{2}x^4}.$$

Question: What is the complete phase portrait of a double well? (continued)

Therefore

h	level-set geometry
$h < -1/4$	empty
$h = -1/4$	the two center equilibria
$-1/4 < h < 0$	two disjoint periodic ovals
$h = 0$	the saddle together with two homoclinic-orbit closures
$h > 0$	one periodic oval surrounding all three equilibria

The two nonconstant homoclinic trajectories on $h = 0$ are not periodic because approach to the saddle takes infinite time.

Visualization: Energy surface for the double well

For

$$w = E(x, y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4,$$

the critical points and heights are

(x, y)	$E(x, y)$	surface type
$(-1, 0)$	$-1/4$	minimum
$(1, 0)$	$-1/4$	minimum
$(0, 0)$	0	saddle

Horizontal slices satisfy

$h < -1/4$	empty
$h = -1/4$	the two minima
$-1/4 < h < 0$	two ovals
$h = 0$	figure-eight saddle-energy level
$h > 0$	one outer oval

and project to the corresponding phase-plane energy levels.

Definition: Heteroclinic trajectory

Let $U \subseteq \mathbb{R}^2$ be open, let $\mathbf{F} : U \rightarrow \mathbb{R}^2$, and let $p, q \in U$ be distinct equilibria of $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$. A solution $\mathbf{z} : \mathbb{R} \rightarrow U$ is **heteroclinic from p to q** if

$$\lim_{t \rightarrow -\infty} \mathbf{z}(t) = p, \quad \lim_{t \rightarrow \infty} \mathbf{z}(t) = q.$$

Its image is a **heteroclinic trajectory** or **heteroclinic orbit**.

Observation: Saddle connections are not periodic

Every homoclinic or heteroclinic solution has an equilibrium limit as $t \rightarrow \infty$. A periodic solution that has a limit as $t \rightarrow \infty$ must be constant: periodicity repeats every point on its trajectory arbitrarily far in the future, so every such point must equal the limit. A homoclinic solution is nonconstant by definition, and a heteroclinic solution has two distinct limiting equilibria. Therefore neither trajectory is periodic.

Recall: Involution

Let U be a set. A map $R : U \rightarrow U$ is an **involution** if

$$R(R(\mathbf{z})) = \mathbf{z}.$$

Recall: Reversing symmetry

Let $U \subseteq \mathbb{R}^2$ be open, let $\mathbf{F} : U \rightarrow \mathbb{R}^2$, and let $R : U \rightarrow U$ be a differentiable involution. The map R is a **reversing symmetry** for $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$ if

$$\mathbf{F}(R(\mathbf{z})) = -DR(\mathbf{z})\mathbf{F}(\mathbf{z}).$$

Observation: A reverser reverses solution arcs

If R is a reversing symmetry and $\mathbf{z}(t)$ is a solution, then $R(\mathbf{z}(-t))$ is also a solution.

Observation: Reflection-reversibility test

For reflection across the x -axis,

$$R(x, y) = (x, -y).$$

Writing $\mathbf{F} = (f, g)$, reversibility becomes

$$f(x, -y) = -f(x, y), \quad g(x, -y) = g(x, y).$$

Proposition: Undamped one-dimensional mechanics is reversible

Let $I \subseteq \mathbb{R}$ be open, let $m > 0$, and let $F : I \rightarrow \mathbb{R}$ be continuous. The system on $I \times \mathbb{R}$

$$\dot{x} = y, \quad \dot{y} = \frac{1}{m}F(x)$$

is reversible under $R(x, y) = (x, -y)$.

Proof. Here

$$\mathbf{F}_{\text{sys}}(x, y) = \left(y, \frac{1}{m}F(x)\right), \quad DR = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Therefore

$$\mathbf{F}_{\text{sys}}(R(x, y)) = \left(-y, \frac{1}{m}F(x)\right) = -DR\mathbf{F}_{\text{sys}}(x, y).$$

Theorem: Nonlinear centers in reflection-reversible systems

Let $U \subseteq \mathbb{R}^2$ be a reflection-invariant open neighborhood of the origin, and let $\mathbf{F} : U \rightarrow \mathbb{R}^2$ be continuously differentiable. Suppose the origin is an equilibrium, the system is reversible under $R(x, y) = (x, -y)$, and $D\mathbf{F}(0, 0)$ has eigenvalues $\pm i\omega$ with $\omega > 0$. Then every sufficiently small non-equilibrium trajectory is closed; the origin is a nonlinear center.

Proof. Reversibility forces the linearization to have the form

$$J = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}, \quad ab = -\omega^2 < 0.$$

After reversing time if necessary, take $a > 0 > b$. On a punctured neighborhood of the origin, define polar variables by

$$r^2 = x^2 + y^2, \quad x = r \cos \theta, \quad y = r \sin \theta,$$

choosing θ continuously along each solution. Then

$$\dot{\theta} = \frac{xy - yx}{r^2} = \frac{bx^2 - ay^2 + o(r^2)}{r^2} \quad (r \rightarrow 0).$$

Since $a > 0 > b$, choose a disk of radius $R > 0$ and constants $c, C > 0$ such that, for $0 < r \leq R$,

$$\dot{\theta} \leq -c < 0, \quad \|\mathbf{F}(\mathbf{z})\| \leq C\|\mathbf{z}\|.$$

The angular bound also shows that there is no other equilibrium in the disk. An angular change of 2π takes at most $2\pi/c$, provided the trajectory stays in the disk. To justify that, use

$$\left| \frac{d}{dt} \log r \right| = \left| \frac{\mathbf{z} \cdot \mathbf{F}(\mathbf{z})}{r^2} \right| \leq C.$$

Thus, while the trajectory is in the disk,

$$r(t) \leq r(0)e^{Ct} \quad (t \geq 0).$$

If $r(0) < Re^{-2\pi C/c}$, this bound prevents exit before time $2\pi/c$; compactness gives continuation up to that time. The trajectory therefore reaches two consecutive crossings of $y = 0$, separated by an angle of π . Reflecting the arc between the crossings and reversing its arrows produces the continuation from the second crossing back to the first. Uniqueness closes the orbit and prevents self-intersections. The loop surrounds the origin because its angle changes by 2π . Uniqueness also places the original initial point on this same periodic orbit.

Question: Does reversibility turn a linear center into a nonlinear center?

Consider

$$\dot{x} = y - y^3, \quad \dot{y} = -x - y^2.$$

Find the equilibria and classify them.

Solution. The first component is odd in y , and the second is even in y , so $R(x, y) = (x, -y)$ is a reverser. At the origin,

$$D\mathbf{F}(0, 0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

whose eigenvalues are $\pm i$. The reversible-center theorem makes the origin a nonlinear center. The equilibrium equations are

$$y(1 - y^2) = 0, \quad -x - y^2 = 0.$$

Thus the other equilibria are $(-1, 1)$ and $(-1, -1)$. Since

$$D\mathbf{F}(x, y) = \begin{pmatrix} 0 & 1 - 3y^2 \\ -1 & -2y \end{pmatrix},$$

the determinant at either point is $-2 < 0$. Both are saddles.

Question: Where is the homoclinic loop?

Consider

$$\dot{x} = y, \quad \dot{y} = x - x^2.$$

Find an exact energy equation for the homoclinic loop in $x \geq 0$.

Solution. The origin is a saddle, and

$$E(x, y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{3}x^3$$

is conserved because

$$\dot{E} = y(x - x^2) + (-x + x^2)y = 0.$$

The saddle energy is $E = 0$. Hence the upper branch satisfies

$$y = x\sqrt{1 - \frac{2x}{3}}, \quad 0 < x \leq \frac{3}{2},$$

and reaches the x -axis at $(3/2, 0)$. Reflection across the x -axis with time reversal gives the lower branch

$$y = -x\sqrt{1 - \frac{2x}{3}}, \quad 0 < x \leq \frac{3}{2}.$$

The two branches approach the origin as $t \rightarrow \pm\infty$ and form a homoclinic loop. More precisely, the upper branch leaves the origin as $t \rightarrow -\infty$ and reaches $(3/2, 0)$ in finite time; the solution then follows the lower branch and approaches the origin as $t \rightarrow +\infty$.

Question: Does reversibility imply conservation?

Consider

$$\dot{x} = -2 \cos x - \cos y, \quad \dot{y} = -2 \cos y - \cos x$$

on $\mathbb{R}^2$. The vector field is 2π -periodic in each coordinate. Show that the system is reversible but not conservative.

Solution. With $R(x, y) = (-x, -y)$, $DR = -I$, and cosine even,

$$\mathbf{F}(R\mathbf{z}) = \mathbf{F}(\mathbf{z}) = -DR\mathbf{F}(\mathbf{z}),$$

so the system is reversible. In the representative cell $[-\pi, \pi) \times [-\pi, \pi)$, its equilibria are

$$(\sigma\pi/2, \rho\pi/2), \quad \sigma, \rho \in \{-1, 1\}.$$

The Jacobian is

$$D\mathbf{F}(x, y) = \begin{pmatrix} 2 \sin x & \sin y \\ \sin x & 2 \sin y \end{pmatrix}.$$

At $(-\pi/2, -\pi/2)$, the eigenvalues are $-3, -1$, so it is an asymptotically stable node and therefore has an open attracting basin. A conservative system cannot have an open attracting basin. Therefore this reversible system is not conservative.

Visualization: Conservation and reversibility

structure	identity	consequence
first integral	$\nabla E \cdot \mathbf{F} = 0$	trajectories remain on level sets
reverser	$\mathbf{F}(R\mathbf{z}) = -DR(\mathbf{z})\mathbf{F}(\mathbf{z})$	reflected arcs carry reversed time
both	two independent checks	energy geometry plus orbit symmetry

Observation: Purely imaginary eigenvalues are inconclusive

Purely imaginary eigenvalues of the Jacobian do not, by themselves, imply the existence of a first integral, a reversing symmetry, or a nonlinear center.